

LEEDALE AGDM, AB, Canada

WMO: 715650

Lat: 52.5529N

Lon: 114.4725W

Elev: 947

StdP: 90.45

Time Zone: -7.00 (NAM)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-32.3	-28.5	-35.8	0.1	-32.3	-31.7	0.2	-28.3	11.2	-3.0	9.9	-6.3	1.1	160	0.694

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	26.9	17.0	25.2	16.2	23.6	15.5	18.8	24.7	17.6	23.6	16.5	22.0	3.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	13.2	22.3	15.3	12.2	20.5	14.2	11.3	19.2	57.5	24.7	53.6	23.7	49.8	22.0	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.9	8.6	7.4	DB	-37.0	30.4	2.8	1.8	-39.0	31.6	-40.6	32.7	-42.2	33.7	-44.2	34.9	(4)
(5)			WB	-36.9	21.2	2.7	1.6	-38.9	22.3	-40.5	23.2	-42.0	24.1	-44.0	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-10.0	-11.0	-4.7	2.8	9.1	12.8	15.3	13.9	9.2	3.4	-5.2	-11.7
	DBStd	11.22	7.66	7.86	7.18	4.92	3.86	2.79	2.75	2.93	3.97	4.33	7.27	7.42
	HDD10.0	3354	621	589	456	218	63	6	1	4	61	209	456	671
	HDD18.3	5948	879	823	714	466	286	166	101	141	273	464	706	929
	CDD10.0	457	0	0	0	3	36	91	164	123	37	3	0	0
	CDD18.3	9	0	0	0	0	0	1	6	2	0	0	0	0
	CDH23.3	410	0	0	0	1	38	45	169	113	41	3	0	0
	CDH26.7	59	0	0	0	0	4	4	30	17	4	0	0	0
(14) Wind	WSAvg	3.0	2.9	2.9	3.2	3.7	3.7	3.4	2.8	2.5	2.6	2.9	2.8	2.7
(15) Precipitation	PrecAvg	539	24	18	23	28	65	101	101	73	43	24	22	19
	PrecMax	669	61	38	48	79	108	184	193	118	91	64	65	42
	PrecMin	411	2	2	2	5	25	22	31	24	5	8	7	5
	PrecStd	74	13	9	10	16	22	35	36	26	23	14	13	10
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	7.6	11.5	22.1	26.9	26.7	29.5	28.2	27.2	21.2	14.0	7.1
		MCWB	3.2	2.8	5.2	10.5	13.1	16.9	19.0	17.3	15.5	11.7	6.8	2.2
	2%	DB	6.1	4.2	8.4	18.4	24.1	24.6	27.1	26.2	24.2	17.1	9.6	3.6
		MCWB	2.1	0.8	3.7	8.7	12.2	15.4	18.6	17.3	14.6	9.4	4.5	0.4
	5%	DB	3.4	2.2	6.5	14.9	21.6	22.5	25.1	24.3	21.6	14.4	5.9	1.1
		MCWB	0.6	-0.3	2.5	6.5	11.1	14.4	17.7	16.7	13.6	7.9	2.4	-1.2
	10%	DB	0.7	0.1	4.5	11.9	19.0	20.5	23.3	22.2	18.5	11.5	3.6	-1.1
		MCWB	-1.3	-1.7	1.3	5.2	10.0	13.3	16.6	15.9	12.2	6.4	1.1	-2.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.2	6.0	10.6	14.2	18.4	21.2	19.6	16.7	12.3	7.5	2.7
		MCDB	7.6	7.3	10.5	21.3	24.7	24.8	26.7	25.0	24.6	20.5	13.2	6.2
	2%	WB	2.2	1.2	4.1	9.0	12.7	16.5	19.5	18.3	15.1	9.8	4.8	0.6
		MCDB	5.8	3.8	8.1	17.9	21.8	22.4	25.4	24.3	22.4	16.3	8.7	3.0
	5%	WB	0.8	-0.2	2.7	7.1	11.6	15.2	18.2	17.2	13.9	8.3	2.9	-1.0
		MCDB	3.3	1.9	6.0	13.8	20.0	20.9	24.2	23.1	20.6	13.4	5.5	1.1
	10%	WB	-1.2	-1.6	1.4	5.6	10.6	14.1	17.1	16.2	12.4	6.8	1.2	-2.7
		MCDB	0.8	0.2	4.1	11.0	18.0	19.4	22.5	21.7	18.1	11.0	3.1	-1.0
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	13.0	12.0	12.6	14.5	12.7	13.6	14.3	14.4	13.6	11.2	11.4
		MCDBR	14.4	14.4	14.0	19.6	20.7	16.9	17.3	18.9	21.4	18.6	14.1	13.3
		MCWBR	10.9	11.3	9.7	10.6	9.9	8.8	9.6	10.9	12.0	11.0	9.6	10.3
	5% WB	MCDBR	14.0	13.7	13.1	18.0	18.1	14.9	16.2	17.2	19.3	17.3	13.8	13.3
		MCWBR	10.6	10.9	9.3	10.1	9.1	8.3	9.4	10.3	11.0	10.6	9.7	10.7
(40) Clear-Sky Solar Irradiance	taub		0.254	0.259	0.299	0.328	0.342	0.349	0.348	0.357	0.311	0.281	0.252	0.241
	taud		2.344	2.414	2.393	2.374	2.385	2.387	2.427	2.371	2.483	2.558	2.489	2.343
	Ebn at Noon		762	880	899	909	909	902	897	867	876	845	776	714
	Edh at Noon		58	73	92	107	112	113	107	107	84	63	50	48
(44) All-Sky Solar Radiation	RadAvg		1.00	1.93	3.33	4.59	5.42	5.75	6.10	5.07	3.56	2.13	1.12	0.77
	RadStd		0.06	0.13	0.20	0.37	0.38	0.43	0.38	0.33	0.38	0.22	0.09	0.05

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page